SCIENTIFIC PORTFOLIO

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Medical Microbiology · Bioinformatics · Biomedical Data Analytics · Data Engineering

A reproducible portfolio connecting medical-biotechnology training and hospital-laboratory experience with computational analysis of biological data.

Research direction

Antimicrobial resistance, bacterial genomics, inflammatory biomarkers, data quality and reproducible biomedical research.

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1. Profile and motivation

I am a Beninese medical biotechnologist and Biomedical Analysis Health Inspector working at CHU-MEL in Cotonou. My practical experience includes haematology, clinical biochemistry, serology/immunology, microbiology, parasitology, antimicrobial susceptibility testing, quality control, biosafety and traceability.

My undergraduate thesis examined inflammatory biomarkers in prostatic dysfunction in a study of 81 patients, including CRP, PSA, leukocytes and cytokines. The thesis received 17.67/20. This portfolio extends that research orientation toward reproducible computational workflows.

Four connected skill areas

• **Clinical microbiology:** understand organism–antibiotic observations and preserve S/I/R categories.

• **Bioinformatics:** perform basic sequence quality checks and summarize FASTA records.

• **Data analyst:** clean data, quantify uncertainty, create visual summaries and explain findings.

• **Data engineering:** ingest, validate, transform, load and trace data with SQL and database workflows.

Data integrity notice

The bundled data are synthetic fixtures for software testing. They are not patient records, not CHU-MEL data and not NCBI records. A real NCBI export was not downloaded in the build environment because outbound network access was unavailable. Therefore, no real-data result is claimed in this edition.

2. Project portfolio

Project 1 — AMR microbiology surveillance

Methods/tools: Python, pandas, quality-control rules, descriptive summaries

What it does: Checks required fields, duplicate isolate–antibiotic records and valid S/I/R categories. Produces counts, resistance proportions and plots while keeping tested denominators visible.

Limit: Key limitation: a proportion among available records is not automatically a population prevalence; source selection and testing standards can bias comparisons.

Project 2 — bacterial sequence bioinformatics

Methods/tools: Biopython, FASTA, sequence length, GC%, ambiguous bases, exploratory ORFs

What it does: Summarizes sequence quality indicators and exports a machine-readable QC table.

Limit: Key limitation: a simple ORF candidate is not a validated gene; these metrics do not replace assembly QC, contamination checks, taxonomy or annotation.

Project 3 — SQL and data engineering

Methods/tools: Relational model, SQLite portable execution, optional DuckDB, ETL, SHA-256 provenance

What it does: Validates schema and values, detects duplicate tests, produces a curated table, builds an analytical database and exports SQL summaries. QC failures stop the pipeline instead of silently changing data.

Limit: Key limitation: a checksum supports file identity, not source truth or scientific correctness.

Project 4 — advanced biomedical data analysis

Methods/tools: Wilson 95% confidence intervals, stratified summaries, isolate–antibiotic matrix, data audit

What it does: Generates grouped resistance estimates, uncertainty intervals, a matrix in which untested results remain missing, and optional year summaries.

Limit: Key limitation: confidence intervals do not correct selection bias, repeated-isolate dependence or inconsistent breakpoint standards.

3. Demonstration run — results and interpretation

All numbers in this section are outputs from the synthetic demonstration only. They are included to show that the code executes and produces interpretable tables, not to make a clinical claim.

Advanced AMR analysis — synthetic fixture

Organism	Antibiotic	Tested	R / I / S	R %	Wilson 95% CI
Escherichia coli	Ceftriaxone	2	1 / 1 / 0	50.0%	9.5–90.5%
Escherichia coli	Ciprofloxacin	2	1 / 0 / 1	50.0%	9.5–90.5%
Escherichia coli	Gentamicin	2	0 / 0 / 2	0.0%	0.0–65.8%
Klebsiella pneumoniae	Ceftriaxone	2	1 / 0 / 1	50.0%	9.5–90.5%
Klebsiella pneumoniae	Ciprofloxacin	2	1 / 0 / 1	50.0%	9.5–90.5%
Klebsiella pneumoniae	Gentamicin	2	1 / 1 / 0	50.0%	9.5–90.5%

Each organism–antibiotic cell has only two synthetic tests, so the intervals are extremely wide. That is the intended lesson: a percentage without its denominator and uncertainty can be misleading. The synthetic examples are too small to support epidemiological conclusions.

Sequence QC — synthetic fixture

Sequence	Length bp	GC %	Ambiguous bases	Interpretation
DEMO_SYNTHETIC_seq01	64	50.00	0	QC indicator only
DEMO_SYNTHETIC_seq02	63	49.21	0	QC indicator only
DEMO_SYNTHETIC_seq03	64	42.86	8	QC indicator only

4. Data engineering and reproducibility

The ETL pipeline reads a CSV, checks required columns, standardizes S/I/R case, records row-level quality issues, stops on ambiguous data, writes a curated table and builds an SQLite analytical database. It also stores a SHA-256 checksum, source URL, query, retrieval date and run timestamp in a provenance manifest.

Recommended run sequence

1. Create a virtual environment and install requirements.
2. Run the AMR Python script and the FASTA QC script.
3. Run the advanced descriptive analysis.
4. Run the SQL loader and ETL pipeline.
5. Inspect the CSV reports, JSON audit/manifests, database and figures.
6. Re-run with a real public export only after documenting its source and mapping columns explicitly.

Example commands

```
python project_01_antimicrobial_resistance/src/analyze_amr.py --input
project_01_antimicrobial_resistance/data/demo/DEMO_SYNTHETIC_amr.csv --outdir
project_01_antimicrobial_resistance/reports/demo_run
```

```
python project_02_bacterial_sequence_bioinformatics/src/sequence_qc.py --fasta
project_02_bacterial_sequence_bioinformatics/data/demo/DEMO_SYNTHETIC_sequences.fasta --outdir
project_02_bacterial_sequence_bioinformatics/reports/demo_run
```

```
python project_04_advanced_biomedical_data_analysis/src/advanced_amr_analysis.py --input
project_03_sql_advanced_biological_data/data/demo/DEMO_SYNTHETIC_amr_sql.csv --outdir
project_04_advanced_biomedical_data_analysis/reports/demo_run --min-tests-per-cell 1
```

```
python project_03_sql_advanced_biological_data/src/etl_pipeline.py --input
project_03_sql_advanced_biological_data/data/demo/DEMO_SYNTHETIC_amr_sql.csv --outdir
project_03_sql_advanced_biological_data/reports/etl_demo --source-url synthetic-fixture
```

5. Public data and sources

NCBI Pathogen Detection documents an Isolates Browser that can integrate isolate metadata, AMR genotypes and AST phenotypes. The AST resource states that phenotype data are submitted into BioSample records, are not available for every isolate and may differ from genotypic resistance calls. The method and coverage must be checked before any comparison.

Official sources

- https://www.ncbi.nlm.nih.gov/pathogens/
- https://www.ncbi.nlm.nih.gov/pathogens/pathogens_help/
- https://www.ncbi.nlm.nih.gov/pathogens/antimicrobial-resistance/resources/
- https://www.ncbi.nlm.nih.gov/pathogens/ast
- https://www.ncbi.nlm.nih.gov/sra

6. Limitations and responsible interpretation

This portfolio is a computational learning and research portfolio, not a validated clinical decision-support system. Results should not guide patient treatment. Public data may be nonrepresentative, unevenly sampled and incomplete. The same isolate may appear more than once or be related to a cluster. Testing methods and breakpoint standards may differ. Missing AST results must not be interpreted as susceptibility. Genotype and phenotype are complementary but non-equivalent evidence.

The current local examples are synthetic. Although the NCBI documentation was consulted, no real NCBI CSV export was downloaded in this build environment because outbound network access was unavailable. The project therefore does not claim real-data results. The ETL is ready to run on a real export after retrieval, column mapping and provenance recording.

7. How to publish and cite on the CV

Publish the repository on GitHub, confirm that the README and all relative links work, and keep confidential or patient-identifiable information out of the repository. Add the public repository URL to the CV and optionally include a direct PDF link. Do not insert a placeholder or untested URL. A suitable CV line is: *Scientific portfolio — Clinical microbiology, bioinformatics, biomedical data analytics & data engineering: [public GitHub URL]*.

The repository includes a step-by-step publication guide in PUBLISH_ON_GITHUB_FR.md and explanatory notes in EXPLANATORY_NOTES_FR.md.